UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 015 · Applications of LDEs: simple harmonic oscillation · compiled 17 October 2026

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1 Short notes

1 · The spring–mass model and simple harmonic motion

Spring–mass: $mg = ks$ at equilibrium; Hooke $-k(s+x)$; Newton $m\ddot{x} = -kx$, so $\ddot{x} + \omega^2 x = 0$, $\omega = \sqrt{k/m}$. Roots $\pm i\omega$: $x = c_1\cos\omega t + c_2\sin\omega t = A\cos(\omega t + \varphi)$, $A = \sqrt{c_1^2 + c_2^2}$, $\tan\varphi = -\frac{c_2}{c_1}$. $T = \frac{2\pi}{\omega} = 2\pi\sqrt{m/k}$, $f = \frac{1}{T}$, $v_{\max} = A\omega$.

2 · Solving spring problems

Recipe: $m = W/g$, $k = W/s$ (feet with $g = 32$, metres with $g = 9.8$); $c_1 = x(0)$, $c_2 = \dot{x}(0)/\omega$ with down positive. 12 lb, 6 in, 4 in down, 2 ft/s up: $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$, $A = \frac{5}{12}$ ft, $T = \frac{\pi}{4}$ s. 8 lb, $k = 12$, raised 5 in: $x = -\frac{5}{12}\cos 4\sqrt{3}t$, $v_{\max} = \frac{5}{\sqrt{3}}$. CW P37: $x = 0.2\cos 14t$, $v_{\max} = 2.8$ m/s, $T = \frac{\pi}{7}$.

3 · The damped oscillator

Damped: $\ddot{x} + 2\lambda n\dot{x} + n^2 x = 0$; $0 < \lambda < 1$: roots $-\lambda n \pm in\sqrt{1-\lambda^2}$, decaying oscillation; from $x(0) = 0$, $\dot{x}(0) = x_0$: $x = \frac{x_0}{n\sqrt{1-\lambda^2}} e^{-\lambda n t}\sin(n\sqrt{1-\lambda^2}\,t)$. $\lambda \geq 1$: no oscillation.

2 Keyword index

20 terms; the number in brackets is the topic.

Amplitude (1). $\sqrt{c_1^2 + c_2^2}$; maximum displacement.

Angular frequency (1). $\omega = \sqrt{k/m}$.

Critical damping (3). $\lambda = 1$; no oscillation.

Damped frequency (3). $n\sqrt{1 - \lambda^2}$.

Damping force (3). $\propto$ velocity, opposes motion.

Downward positive (2). Sign convention for x and $\dot{x}$.

Equilibrium position (1). $mg = ks$; $x = 0$.

Free motion (2). No damping, no external force.

Hooke's law (1). Restoring force $-k(s + x)$.

Maximum velocity (1). $A\omega$, at equilibrium.

Natural frequency (1). $\frac{1}{T} = \frac{\omega}{2\pi}$.

Newton's second law (1). Net force $= m\ddot{x}$.

Overdamped (3). $\lambda > 1$; real roots.

Period (1). $2\pi\sqrt{m/k}$.

Phase (1). $\tan\varphi = -c_2/c_1$.

Released (2). $\dot{x}(0) = 0$.

Simple harmonic motion (1). $x = A\cos(\omega t + \varphi)$.

Slug (2). Mass unit: lb ÷ 32 ft/s².

Spring constant (1). $k = W/s$.

Underdamped (3). $0 < \lambda < 1$; decaying oscillation.

3 Rapid recall

1	The two forces on a hanging mass?	Gravity mg and the restoring force $-k(s+x)$.
2	Equilibrium condition?	$mg = ks$.
3	The equation of SHM?	$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$, i.e. $\ddot{x} + \omega^2 x = 0$.
4	Its general solution?	$c_1 \cos \omega t + c_2 \sin \omega t$.
5	Amplitude from c_1, c_2?	$\sqrt{c_1^2 + c_2^2}$.
6	Period and frequency?	$T = 2\pi\sqrt{m/k}$, $f = \frac{1}{T}$.
7	Maximum velocity?	$A\omega$.
8	g in feet?	32 ft/s².
9	Mass of a 12 lb weight?	$\frac{3}{8}$ slug.
10	k if 12 lb stretches 6 in?	24 lb/ft.
11	Upward velocity 2 ft/s: $\dot{x}(0)$?	-2 (down positive).
12	12 lb example: $x(t)$?	$\frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$.
13	...its amplitude and period?	$\frac{5}{12}$ ft, $\frac{\pi}{4}$ s.
14	8 lb, $k = 12$: ω?	$4\sqrt{3}$ rad/s.
15	CW P37: $x(t)$?	$0.2\cos 14t$ m.
16	Damped equation of slide 015·15?	$\ddot{x} + 2\lambda n\dot{x} + n^2 x = 0$.
17	Its roots for $0 < \lambda < 1$?	$-\lambda n \pm in\sqrt{1-\lambda^2}$.
18	When does a damped mass not oscillate?	$\lambda \geq 1$.

Mnemonic (study aid). *"Weight over g, weight over stretch."* $m = \frac{W}{g}$, $k = \frac{W}{s}$: the two numbers every spring problem needs first (study aid).

4 Last 10 minutes before the exam

Top facts

1. $mg = ks$ at equilibrium; then $\ddot{x} + \frac{k}{m}x = 0$, $\omega = \sqrt{k/m}$.
2. $x = c_1 \cos \omega t + c_2 \sin \omega t = A\cos(\omega t + \varphi)$, $A = \sqrt{c_1^2 + c_2^2}$, $\tan \varphi = -\frac{c_2}{c_1}$.
3. $T = \frac{2\pi}{\omega} = 2\pi\sqrt{m/k}$, $f = \frac{1}{T}$, $v_{\max} = A\omega$.
4. $m = \frac{W}{g}$, $k = \frac{W}{s}$; inches → feet, $g = 32$ ft/s² (or 9.8 m/s²).
5. Down positive: pulled down +, raised −, upward velocity −, released $\dot{x}(0) = 0$.
6. $c_1 = x(0)$, $c_2 = \frac{\dot{x}(0)}{\omega}$.
7. 12 lb: $x = \frac{1}{3}\cos 8t - \frac{1}{4}\sin 8t$, $A = \frac{5}{12}$, $T = \frac{\pi}{4}$. 8 lb: $x = -\frac{5}{12}\cos 4\sqrt{3}t$, $v_{\max} = \frac{5}{\sqrt{3}}$.
8. Damped: $\ddot{x} + 2\lambda n\dot{x} + n^2 x = 0$, $x = \frac{x_0}{n\sqrt{1-\lambda^2}}e^{-\lambda n t}\sin(n\sqrt{1-\lambda^2}t)$.

Top formulas

E1.1 · Equilibrium condition

ESA ★★

$$mg = ks$$

gives	$k = \frac{mg}{s} = \frac{W}{s}$ from a static stretch s.
applies	The mass hangs at rest.
units	W in lb with s in ft (lb/ft); W in N with s in m (N/m).
note	$W = mg$ is the weight.

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E1.3 · Equation of simple harmonic motion

ESA ★★

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0, \qquad \frac{d^2x}{dt^2} + \omega^2 x = 0, \quad \omega = \sqrt{\frac{k}{m}}$$

gives	The model of the undamped, unforced spring.
applies	No damping, no external force.
note	A homogeneous LDE with constant coefficients: AE $m^2 + \omega^2 = 0$ (class 002, complex roots).

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E1.4 · General solution

ESA ★★

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

GIVES The position at any instant.

APPLIES Roots $\pm i\omega$.

NOTE $c_1 = x(0)$, $c_2 = \frac{\dot{x}(0)}{\omega}$.

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E1.5 · Amplitude–phase form

ESA ★★

$$x(t) = A\cos(\omega t + \varphi), \quad c_1 = A\cos\varphi,\ c_2 = -A\sin\varphi, \quad A = \sqrt{c_1^2 + c_2^2},\ \tan\varphi = -\frac{c_2}{c_1}$$

GIVES Amplitude and phase from c_1, c_2.

APPLIES After the constants are known.

NOTE Pick the quadrant of φ from the signs of $\cos\varphi = \frac{c_1}{A}$ and $\sin\varphi = -\frac{c_2}{A}$.

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E1.6 · Period and natural frequency

ESA ★★

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}, \qquad f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

GIVES Seconds per oscillation; oscillations per second.

APPLIES Undamped motion.

NOTE The lecturer: “you should remember all these formulae”. The last form of f is added for clarity.

S015·9 · T015 00:07:40 · T015 00:07:58

E1.7 · Maximum velocity

ESA ★★

$$\dot{x} = -A\omega\sin(\omega t + \varphi), \qquad v_{\max} = A\omega$$

GIVES The greatest speed, reached at the equilibrium position.

APPLIES Any SHM.

NOTE Maximum acceleration is $A\omega^2$ (added for clarity).

S015·22 · CW U2 P37

E2.2 · Constants from the initial conditions ESA ★★

$$c_1 = x(0), \qquad c_2 = \frac{\dot{x}(0)}{\omega}$$

GIVES The particular motion.

APPLIES Downward positive: pulled down d → $x(0) = +d$; raised d → $x(0) = -d$; upward velocity v → $\dot{x}(0) = -v$; released → $\dot{x}(0) = 0$.

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E3.3 · Damped motion from $x(0) = 0$, $\dot{x}(0) = x_0$ ESA ★

$$x(t) = \frac{x_0}{n\sqrt{1-\lambda^2}}\, e^{-\lambda n t} \sin\!\left(n\sqrt{1-\lambda^2}\, t\right)$$

GIVES The slide's boxed answer.

APPLIES $0 < \lambda < 1$.

NOTE Checked by substitution and both initial conditions (sympy).

S015·16